



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 4: Probability IV

evans.gouno@univ-ubs.fr

1 Bivariate random variables

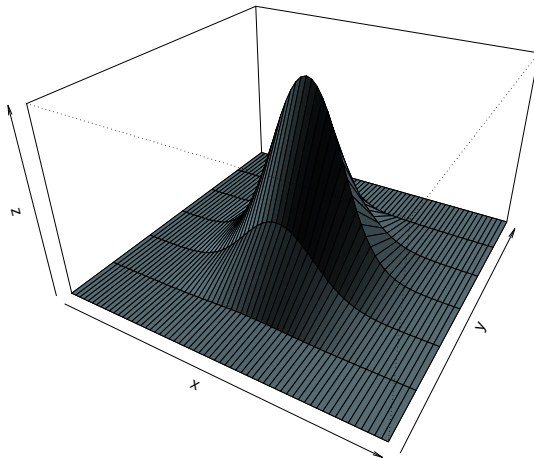
2 Independent r.v.'s

Couple of r.v.'s

- Distribution of a bivariate random variables (joint distribution):
For $(x, y) \in U \times V$,
 - $P(X = x, Y = y)$, if X and Y are discrete r.v.'s.
 - $f_{X,Y}(x, y)$, if X and Y are continuous r.v.'s (bivariate p.d.f.)
- Conditions:
 - $\sum_{(x,y)} P(X = x, Y = y) = 1$ (discrete),
 - $\int_{U \times V} f_{(X,Y)}(x, y) dx dy = 1$ (continuous).

Bivariate random variable

Example of a bivariate r.v. representation



● Marginal distributions

- $P(X = x) = \sum_y P(X = x, Y = y)$ and $P(Y = y) = \sum_x P(X = x, Y = y)$
(discrete)

- $f_X(x) = \int f_{(X,Y)}(x,y)dy$ and $f_Y(y) = \int f_{(X,Y)}(x,y)dx$ (continuous).

● Conditional distribution

- $P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$ (discrete)

- $f_{X|Y=y}(x) = \frac{f_{(X,Y)}(x,y)}{f_Y(y)}$ (Continuous version of the Bayes's theorem).

Expectation and Covariance



$$E[g(X, Y)] = \begin{cases} \sum_{x,y} g(x,y)P(X=x, Y=y) & \text{if } X \text{ and } Y \text{ are discrete,} \\ \int g(x,y)f_{X,Y}(x,y)dxdy & \text{if } X \text{ and } Y \text{ are continuous.} \end{cases}$$

Expectation and Covariance



$$E[g(X, Y)] = \begin{cases} \sum_{x,y} g(x,y)P(X=x, Y=y) & \text{if } X \text{ and } Y \text{ are discrete,} \\ \int g(x,y)f_{X,Y}(x,y)dx dy & \text{if } X \text{ and } Y \text{ are continuous.} \end{cases}$$

- $E(X + Y) = EX + EY$

Expectation and Covariance



$$E[g(X, Y)] = \begin{cases} \sum_{x,y} g(x,y)P(X=x, Y=y) & \text{if } X \text{ and } Y \text{ are discrete,} \\ \int g(x,y)f_{X,Y}(x,y)dxdy & \text{if } X \text{ and } Y \text{ are continuous.} \end{cases}$$

- $E(X + Y) = EX + EY$
- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

Expectation and Covariance



$$E[g(X, Y)] = \begin{cases} \sum_{x,y} g(x,y)P(X=x, Y=y) & \text{if } X \text{ and } Y \text{ are discrete,} \\ \int g(x,y)f_{X,Y}(x,y)dxdy & \text{if } X \text{ and } Y \text{ are continuous.} \end{cases}$$

- $E(X + Y) = EX + EY$
- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$
- The **covariance** of two random variables is:

$$Cov(X, Y) = E[(X - EX)(Y - EY)] = E(XY) - E(X)E(Y).$$

Expectation and Covariance



$$E[g(X, Y)] = \begin{cases} \sum_{x,y} g(x,y)P(X=x, Y=y) & \text{if } X \text{ and } Y \text{ are discrete,} \\ \int g(x,y)f_{X,Y}(x,y)dxdy & \text{if } X \text{ and } Y \text{ are continuous.} \end{cases}$$

- $E(X + Y) = EX + EY$
- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$
- The **covariance** of two random variables is:

$$Cov(X, Y) = E[(X - EX)(Y - EY)] = E(XY) - E(X)E(Y).$$

- **Remark:**

$$Cov(X, X) = E[(X - EX)(X - EX)] = E[(X - EX)^2] = Var(X).$$

Independent r.v.'s

Definition

Two random variables are said to be independent if the distribution of the couple is the product of each variable distribution.

$$(X \text{ and } Y \text{ independent}) \iff \begin{cases} P(X=x, Y=y) = P(X=x)P(Y=y), \\ \text{for discrete r.v.'s.} \\ \\ f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y), \\ \text{for continuous r.v.'s.} \end{cases}$$

Independent r.v.'s

Exercise – Let $f_{X,Y}(x,y) = 4(1-x)y$, for $x,y \in [0,1]$
and $f_{X,Y}(x,y) = 0$ otherwise.

- ➊ Show that $f(x,y)$ is a joint probability distribution function.
- ➋ Find the marginal p.d.f. of X and Y .
- ➌ Are X and Y independent?
- ➍ Find the probability $P(X \leq 1/2 \mid Y > 1/3)$.

Independent r.v.'s

Proposition

Let X and Y be two r.v.'s

If X and Y are independent then $\text{Cov}(X, Y) = 0$.

Remark: $\text{Cov}(X, Y) = 0 \iff E(XY) = E(X)E(Y)$.

Warning: $\text{Cov}(X, Y) = 0$ does not imply that X, Y are independent.

Exercise:

Let (X, Y) be a bivariate r.v. taking values in $V = \{(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)\}$ such that $P(X = i, Y = j) = 1/5, \forall (i, j) \in V$.

- 1 Are X and Y independent?
- 2 Compute $\text{Cov}(X, Y)$.

Independent r.v.

Proposition

Let $X_1, X_2, \dots, X_n$ be n independent r.v. Denote $S_n = \sum_{i=1}^n X_i$.

Then: $M_{S_n}(t) = \prod_{i=1}^n M_{X_i}(t)$.

Proof: $M_{S_n}(t) = E(e^{tS_n}) = E(e^{t\sum_{i=1}^n X_i}) = E\left(\prod_{i=1}^n e^{tX_i}\right)$.

Since the r.v. are independent: $E\left(\prod_{i=1}^n e^{tX_i}\right) = \prod_{i=1}^n E(e^{tX_i})$.

Therefore $M_{S_n}(t) = \prod_{i=1}^n M_{X_i}(t)$.